

DETAILED DESCRIPTION OF THE INVENTION — SOFTWARE ARCHITECTURE, SIGNAL PROCESSING PIPELINES, AND ANALYTIC FRAMEWORKS

The intraluminal physiologic monitoring device may operate in conjunction with software architectures configured to acquire, process, analyze, and interpret physiologic signals generated by onboard sensing systems. The invention therefore encompasses computational frameworks supporting transformation of raw sensor data into actionable physiologic insight.

In various embodiments, software systems may implement multi-stage signal processing pipelines including signal acquisition, filtering, normalization, feature extraction, pattern recognition, and interpretation layers. Processing stages may operate locally on-device, within companion applications, or through distributed cloud computing environments.

Signal conditioning techniques may include noise reduction, motion artifact rejection, baseline correction, frequency-domain analysis, temporal smoothing, adaptive thresholding, or sensor fusion algorithms combining multiple sensing modalities into unified physiologic representations.

Analytic frameworks may employ rule-based analysis, statistical modeling, machine learning algorithms, neural network architectures, probabilistic inference systems, or hybrid computational methods capable of identifying physiologic states, transitions, or predictive trends.

Software architectures may support real-time processing during active measurement sessions as well as retrospective analysis performed after data synchronization. Real-time pipelines may provide immediate feedback while historical analysis may generate longitudinal insights.

Sensor fusion systems may correlate pressure, motion, impedance, temperature, or other physiologic signals to derive higher-order metrics representing complex physiologic behaviors not directly measurable by individual sensors alone.

Certain embodiments may implement adaptive analytics capable of modifying interpretation models based on individual user history, aggregated population data, or contextual environmental inputs. Continuous model refinement may improve accuracy over time.

Software components may be structured as modular services allowing independent updates to analytic algorithms, visualization systems, privacy layers, or communication interfaces without requiring modification of device hardware.

Future analytic environments may integrate artificial intelligence agents, predictive coaching systems, automated anomaly detection engines, or collaborative analysis platforms operating across interconnected physiologic monitoring networks.

The software architectures and analytic frameworks described herein are illustrative rather than limiting. Any computational system capable of processing intraluminal physiologic sensing data into interpretable information is considered within the scope of the present disclosure.